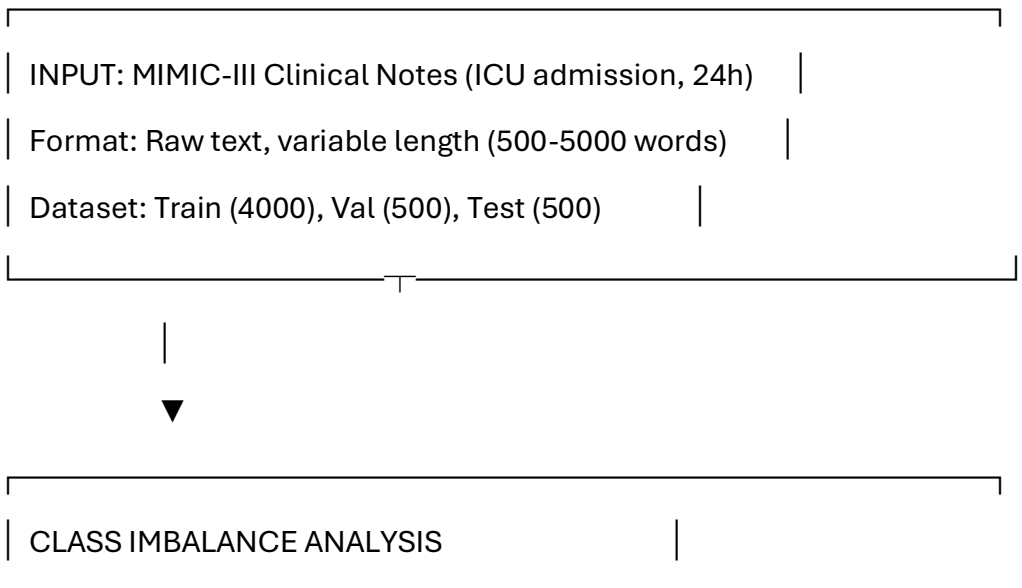
AMIA draft- SFT with Bio_ClinicalBERT, 110M parameters)

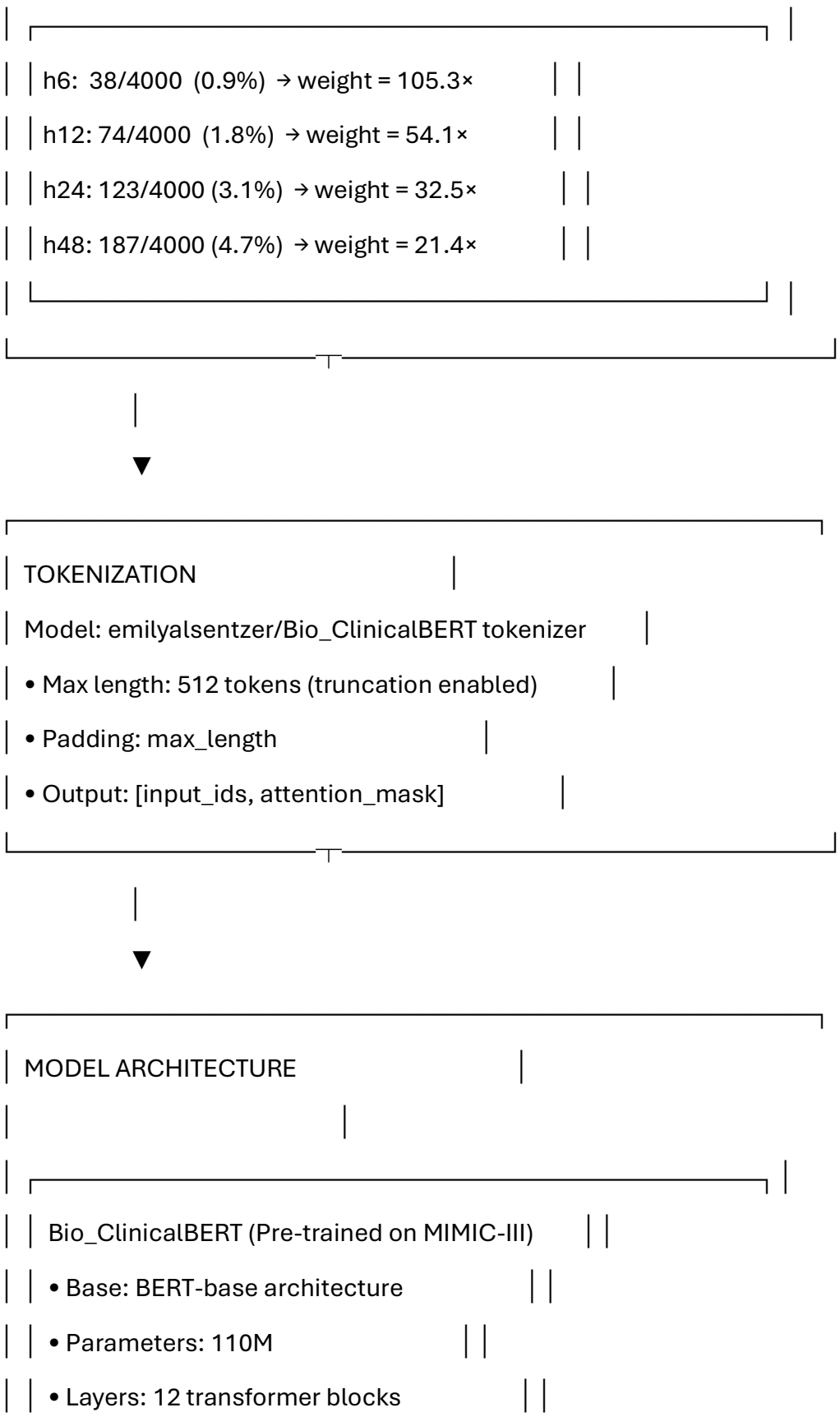
Prepared by: Seraj Mostafa

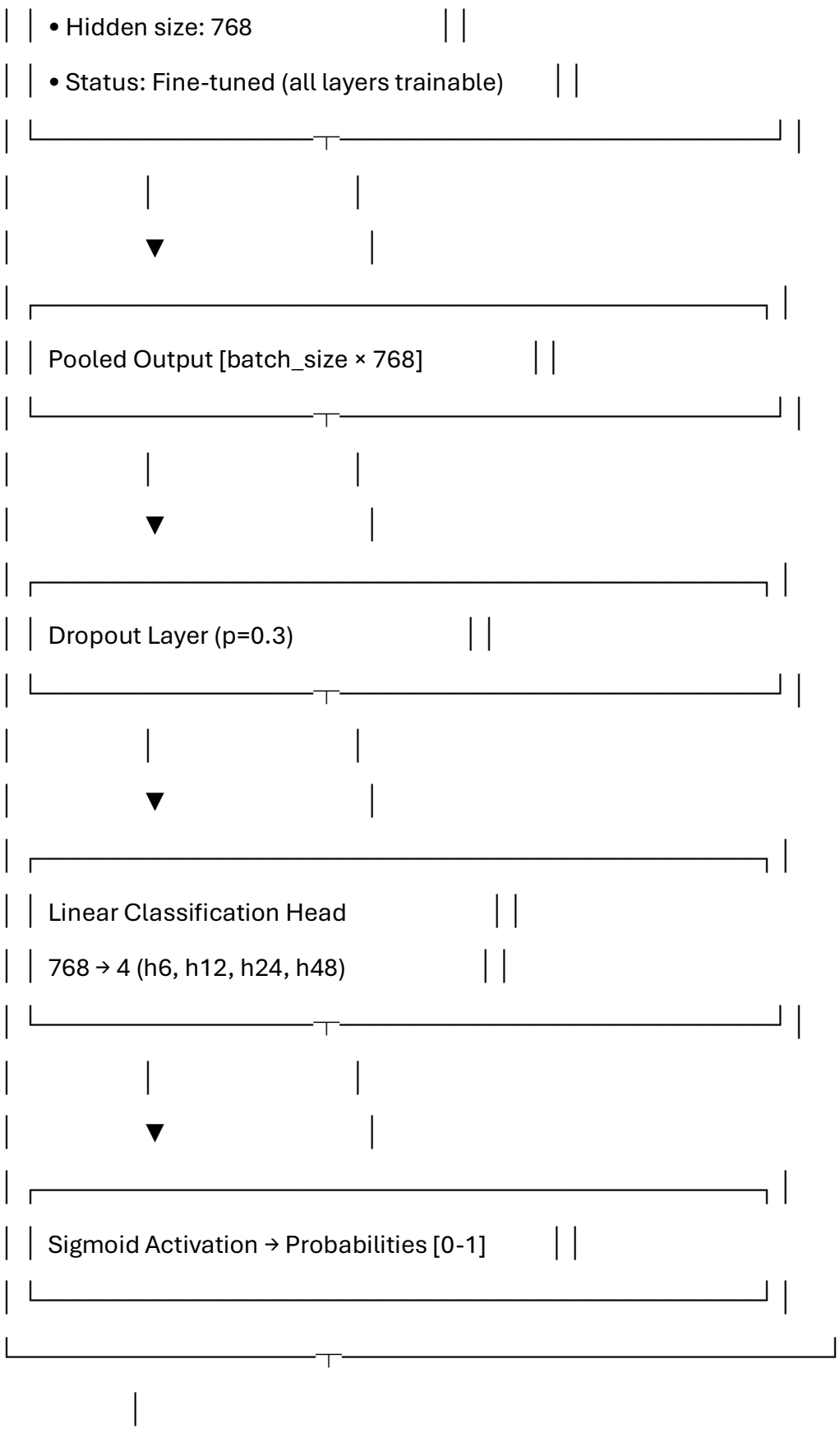
Summary:

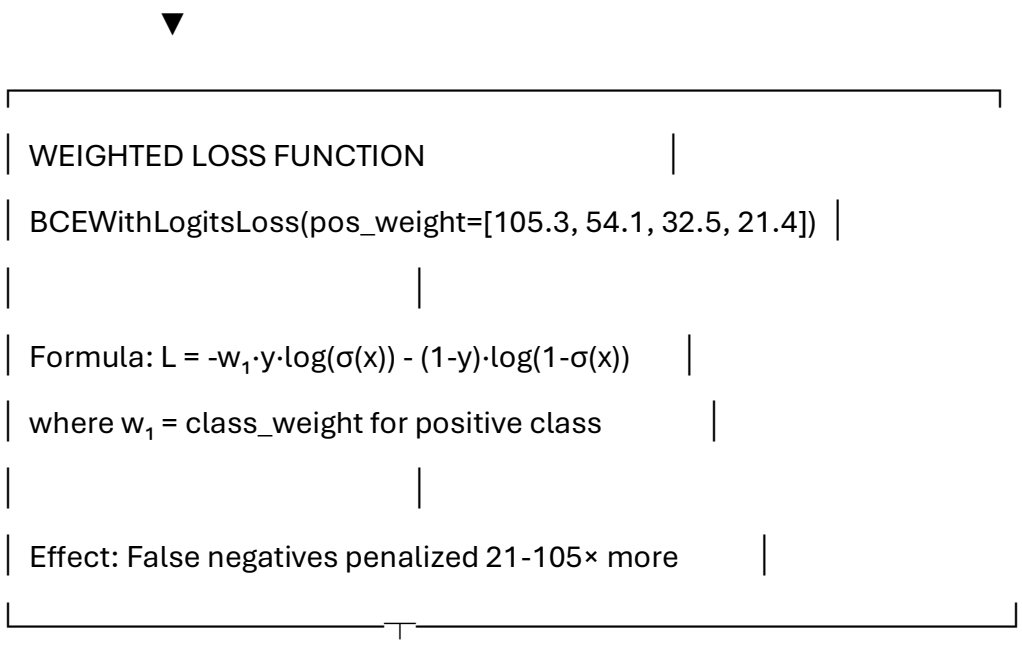
We fine-tuned Bio_ClinicalBERT (emilyalsentzer/Bio_ClinicalBERT, 110M parameters), a BERT-based model pre-trained on MIMIC-III clinical notes, for multi-label sepsis prediction at four time horizons (6, 12, 24, and 48 hours). Given the severe class imbalance in our dataset (0.9%-4.7% positive cases across time points), we employed a weighted binary cross-entropy loss function with dynamically calculated class weights inversely proportional to class frequencies (ranging from 21.3× to 105.3× penalty for false negatives). Clinical notes were tokenized using the ClinicalBERT tokenizer with a maximum sequence length of 512 tokens. We added a dropout layer (p=0.3) and a linear classification head (768→4 dimensions) on top of the frozen BERT encoder's pooled output. The model was trained for 15 epochs using the AdamW optimizer (learning rate: 2×10^{-5}, batch size: 16) with linear warmup scheduling (100 steps) on an T4 GPU. Model selection was based on the highest average F1-score across all four time points on the validation set. Performance was evaluated using recall (sensitivity), precision, F1-score, and geometric mean (G-Mean) to account for class imbalance, with particular emphasis on recall as the primary metric given the clinical imperative to minimize false negatives in sepsis detection.

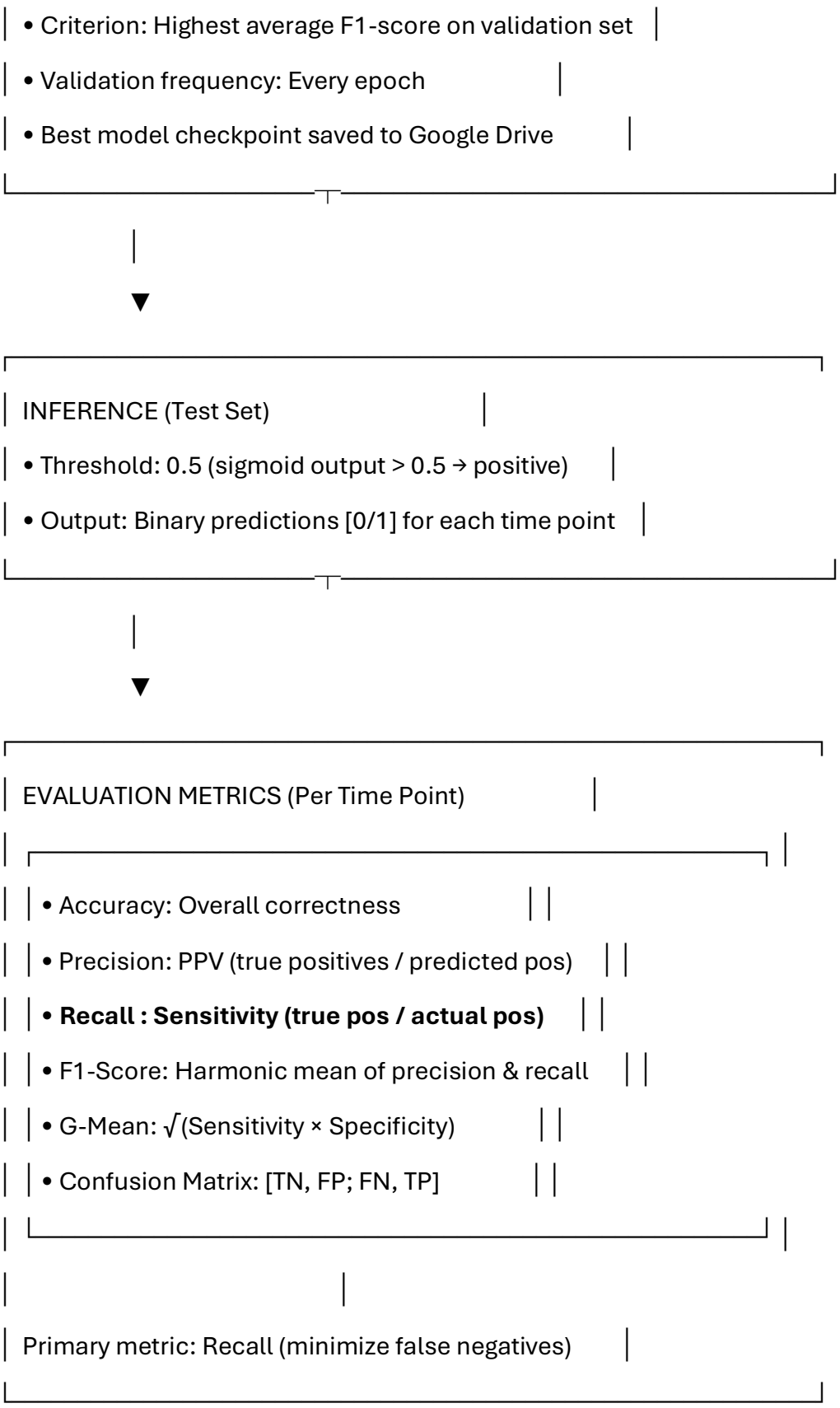
Workflow:




h6: 38/4000 (0.9%) → weight = 105.3×
h12: 74/4000 (1.8%) → weight = 54.1×
h24: 123/4000 (3.1%) → weight = 32.5×
h48: 187/4000 (4.7%) → weight = 21.4×
TOKENIZATION
Model: emilyalsentzer/Bio_ClinicalBERT tokenizer
• Max length: 512 tokens (truncation enabled)
• Padding: max_length
• Output: [input_ids, attention_mask]
MODEL ARCHITECTURE
Bio_ClinicalBERT (Pre-trained on MIMIC-III)
• Base: BERT-base architecture
• Parameters: 110M
• Layers: 12 transformer blocks


• Hidden size: 768
• Status: Fine-tuned (all layers trainable)
Pooled Output [batch_size × 768]
Dropout Layer (p=0.3)
Linear Classification Head
768 → 4 (h6, h12, h24, h48)
Sigmoid Activation → Probabilities [0-1]


WEIGHTED LOSS FUNCTION
BCEWithLogitsLoss(pos_weight=[105.3, 54.1, 32.5, 21.4])
Formula: L = -w1·y·log(σ(x)) - (1-y)·log(1-σ(x))
where w1 = class_weight for positive class
Effect: False negatives penalized 21-105× more
OPTIMIZATION
• Optimizer: AdamW (torch.optim)
• Learning rate: 2×10−5
• Batch size: 16
• Epochs: 15
• Scheduler: Linear warmup (100 steps) + linear decay
• Hardware: NVIDIA A100-SXM4-40GB
• Training time: ~30 minutes
MODEL SELECTION


• Criterion: Highest average F1-score on validation set
• Validation frequency: Every epoch
• Best model checkpoint saved to Google Drive
INFERENCE (Test Set)
• Threshold: 0.5 (sigmoid output > 0.5 → positive)
• Output: Binary predictions [0/1] for each time point
EVALUATION METRICS (Per Time Point)
• Accuracy: Overall correctness
• Precision: PPV (true positives / predicted pos)
• Recall : Sensitivity (true pos / actual pos)
• F1-Score: Harmonic mean of precision & recall
• G-Mean: √(Sensitivity × Specificity)
• Confusion Matrix: [TN, FP; FN, TP]
Primary metric: Recall (minimize false negatives)